

Inferring critical transitions from timeseries

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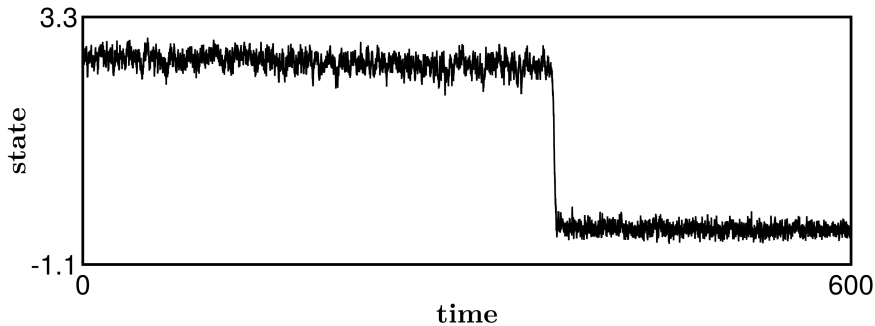
†Auckland Bioengineering Institute



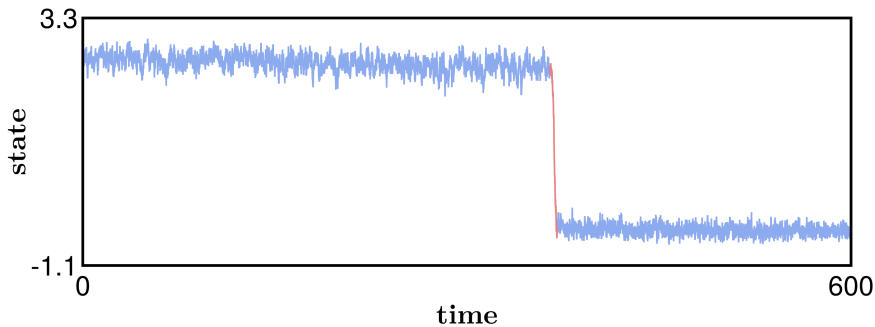
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Taumata Rau
**University
of Auckland**



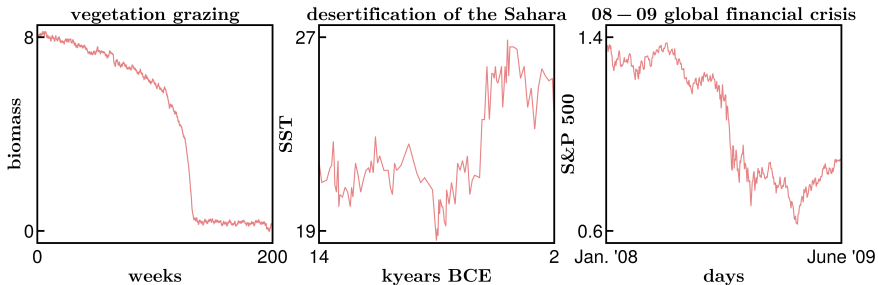
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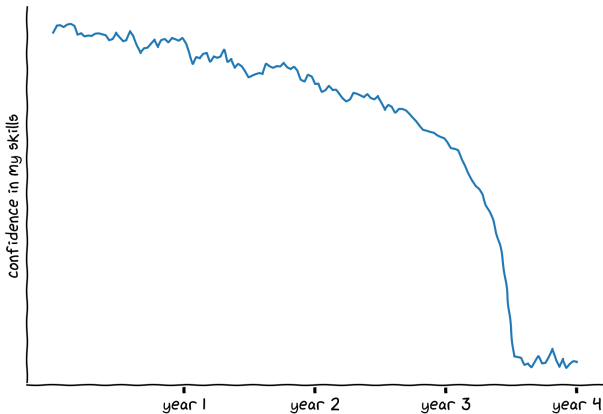


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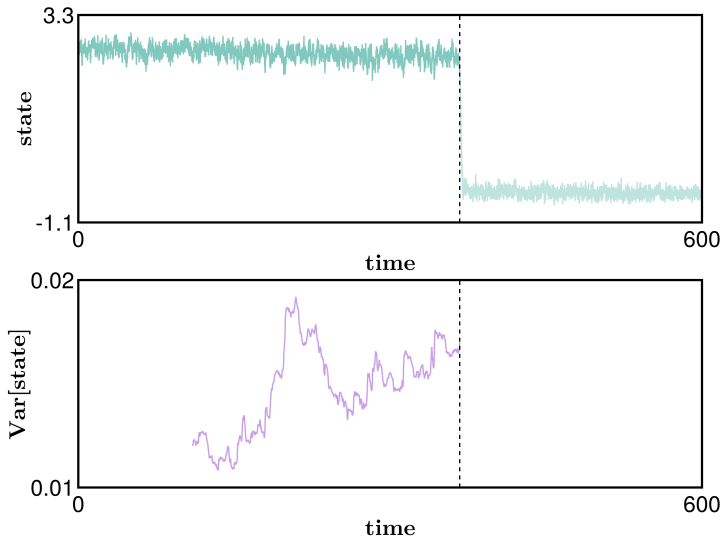


Inferring **critical transitions** from timeseries

A PhD's tipping point in imposter syndrome



Inferring critical transitions from timeseries



The first early-warning signal (EWS)

Consider the following Itô process with additive white noise (i.e. $\sigma > 0$)

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Assuming we start the process at fixed $X_0 = \sqrt{-\mu} + \delta$ ($\delta \ll 1$), a solution in a neighbour of $X_t = \sqrt{-\mu}$ is well approximated by the solution of the linear Ornstein-Uhlenbeck process (OUP)

$$X_t = \theta \int_0^t (\sqrt{-\mu} - X_s) ds + \sigma \int_0^t dW_s,$$

which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

$$\text{Cov}[X_s, X_t] = \frac{\sigma^2}{2\theta} e^{-\theta(t-s)}, \quad s < t.$$

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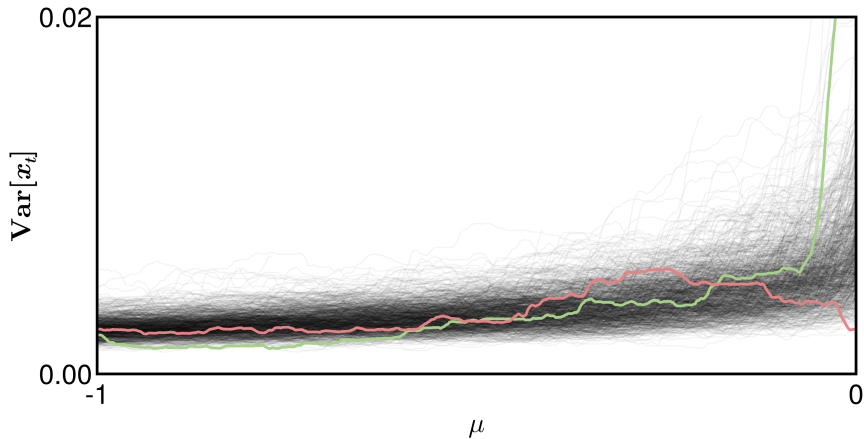
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If we now consider instantaneous time increments (i.e. $t \rightarrow s$) then

$$\text{Cov}[X_s, X_t] \rightarrow \text{Var}[X_t] = \frac{\sigma^2}{2\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$

A quest for robust EWSs



Large deviation theory

Definition (Large deviation principle)

Given a random variable X_t evolving according to $dX_t = f(X_t)dt + \sqrt{\varepsilon}dW_t$, initial condition X_0 and total time $T > 0$, the probability of the sample path $X^\varepsilon = \{X_t\}_{t=0}^T$ being located in a cylinder of radius $\delta > 0$ around a path $\phi(X_0, t)$ satisfies a LDP if

$$P[\phi] \asymp e^{-\frac{S[\phi]}{\varepsilon}} \quad \left(\text{or} \quad \lim_{\varepsilon \rightarrow 0} \log P \left(\sup_{t \in [0, T]} |X_t - \phi(X_0, t)| < \delta \right) = -S[\phi] \right),$$

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Theorem (Freidlin-Wentzell)

The sample paths of $dX_t = f(X_t)dt + \sqrt{\varepsilon}dW_t$ satisfy a LDP with rate functional

$$S[\phi] = \int_0^T L(\dot{\phi}, \phi, t) dt = \frac{1}{2} \int_0^T \left(\dot{\phi} - f(\phi) \right)^2 dt,$$

where ϕ is any (absolutely continuous) deterministic trajectory.

Large deviation theory

The most likely path $\phi(X_0, t) \in H^1(\mathbb{R}^d, [0, T])$ is the minimiser of $S[\phi]$ and satisfies the Euler-Lagrange equations (ELEs)

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{\phi}} L(\dot{\phi}, \phi, t) \right) - \frac{\partial}{\partial \phi} L(\dot{\phi}, \phi, t) = 0.$$

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Example (OUP)

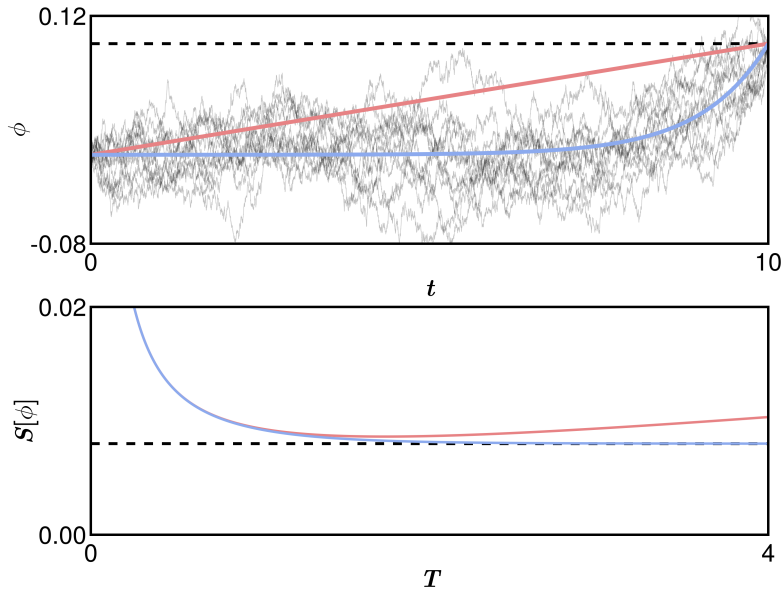
Consider $dX_t = -X_t dt + \sqrt{\varepsilon} dW_t$ with initial $X_0 = 0$ and termination $X_T = Y$ conditions. Given two paths

$$\phi^{(1)}(0, t) = Y \frac{t}{T}, \quad \phi^{(2)}(0, t) = Y \frac{\sinh(t)}{\sinh(T)},$$

with $\phi^{(1)}$ being an arbitrary linear (in time) trajectory connecting 0 to Y and $\phi^{(2)}$ solving the ELEs, their rate functionals are

$$\mathcal{S}^{(1)} := S[\phi^{(1)}] = \frac{Y^2}{2T} \left(1 + T + \frac{T^2}{3} \right), \quad \mathcal{S}^{(2)} := S[\phi^{(2)}] = \frac{Y^2}{4 \sinh^2 T} (e^{2T} - 1).$$

Large deviation theory



Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the LDP for hitting the target x scales like $\exp(V(x_0) - V(x))$.

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Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb,uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

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Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

$$V(b) - V(a) =: \Delta V \rightarrow 0 \implies \exp(-\Delta V) \rightarrow 1.$$

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We can use the LDP as an EWS!

$$\exp(-\Delta V) = \begin{cases} 0, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow \infty, \\ 1, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow 0. \end{cases}$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ is described, over the state space, by the solution of the Fokker-Plank (or forward Kolmogoroff) equation

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Let $\mu \in \mathbb{R}$ be fixed over $\Delta t \gg 0$; if $\frac{\Delta p}{\Delta t} \approx 0$ (i.e. I ignore the transient regime) then the transition probability solves the second-order IVP

$$\begin{cases} Dp'' + V''p + V'p' = 0, \\ p'(\pm\infty) = 0. \end{cases}$$

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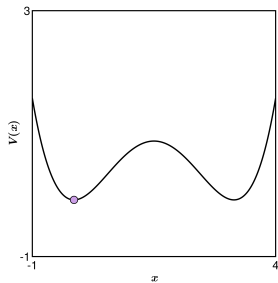
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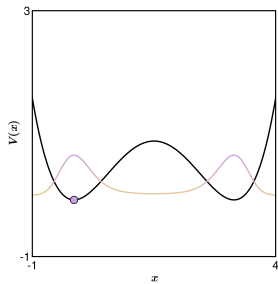
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I can then solve for $p(x)$ using an integration factor and integration-by-parts to get a **stationary equilibrium density** $p(x) = N e^{-\frac{V(x)}{D}}$.

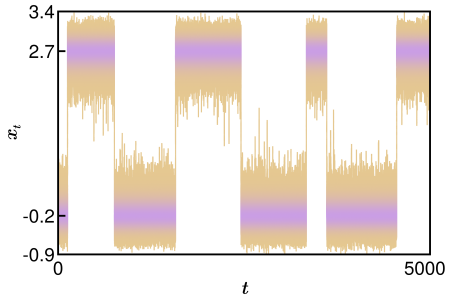
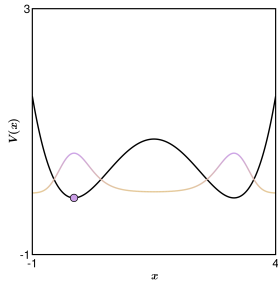
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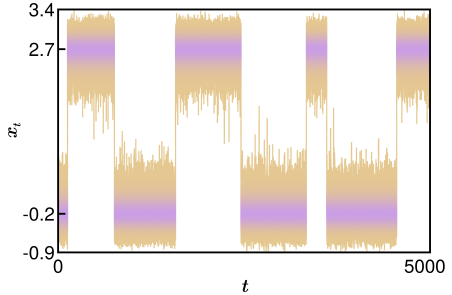
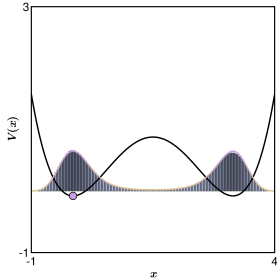
Transition probabilities



Transition probabilities



Transition probabilities



A numerical method to predict tipping

Input: timeseries x_t over $\Delta t > 0$, threshold > 0

define $V(x, \mathbf{c}) = \mathbf{c}[1]x + \mathbf{c}[2]x^2 + \mathbf{c}[3]x^3$;

define $\rho(x, \mathbf{c}) = \exp(-V(x, \mathbf{c}))$;

residuals = detrend(x_t , Δt);

histogram = fit(residuals);

while *error* > *threshold* **do**

 guess = solve_lls(histogram) + rand(3);

$p(x)$ = normalise($\rho(x, \text{guess})$);

 solution = solve_nlls($p(x)$, histogram, guess);

 error = $\int (\rho(x, \text{solution}) - \text{histogram})^2 dx$

a, b = compute_extrema($V(x, \text{guess})$);

ews = $\exp(-(V(b, \text{guess}) - V(a, \text{guess})))$;

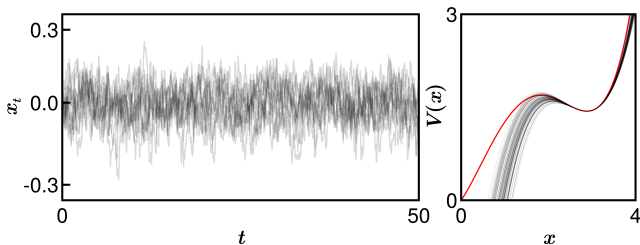
Output: approximate **ews** from the the reconstruction of V

Nonlinear least-squares regression

To characterise the statistical properties of the method let's consider an ensemble of $N \gg 1$ i.i.d. overdamped particles in a cubic potential

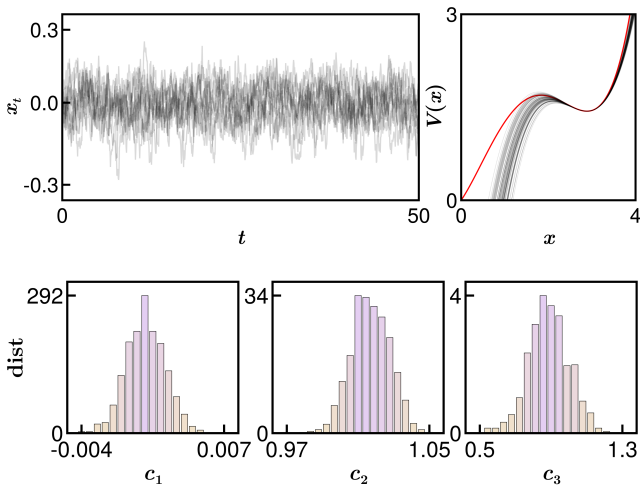
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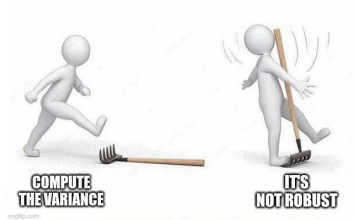


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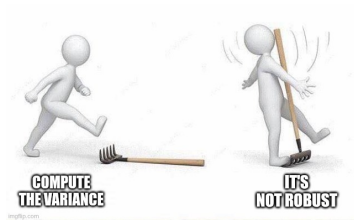
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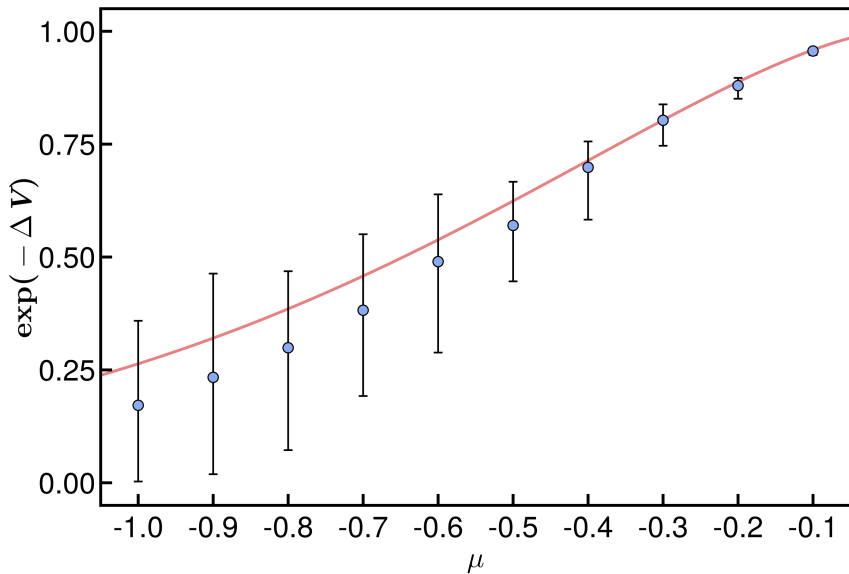
Does it work?



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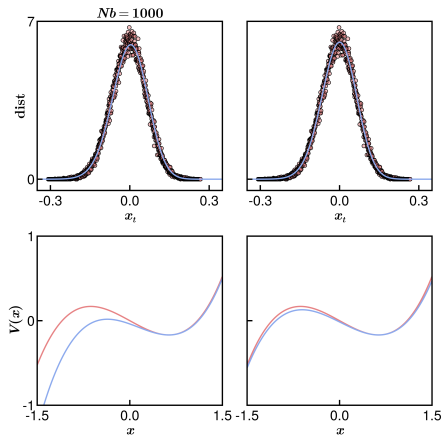
That's the end folks!

-  D. W. Scott, *Biometrika* **66** (1979).
-  C. Kuehn, *Physica D* **240** (2011).
-  J. Lohmann and G. Gottwald, *arXiv* (2025).
-  N. Berglund and B. Gentz, *J. Diff. Eq.* **191** (2003).
-  P. Ashwin, et al., *Nonlinearity* **30** (2017).
-  P. Ritchie and J. Sieber, *Chaos* **26** (2016).
-  J. Shao, T. Grafke, and R. MacKay, *arXiv* (2025).

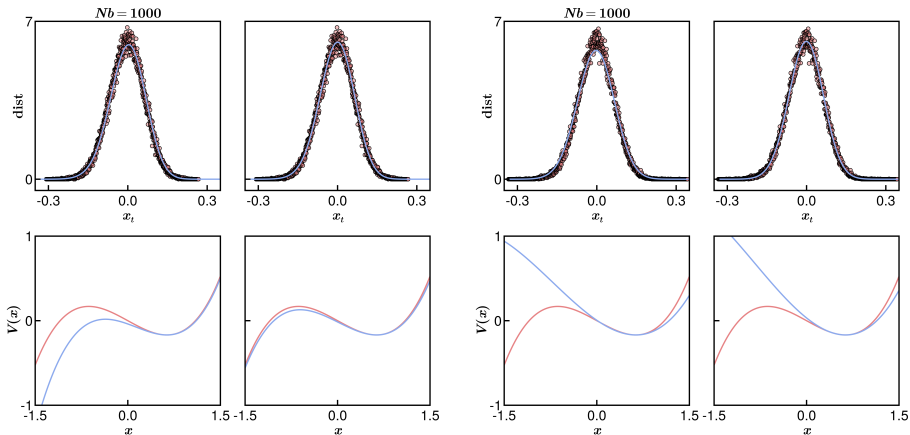
Thank you.

► [Go to the beginning](#)

Investigating the lack of robustness



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Symmetrically weighted least-squares

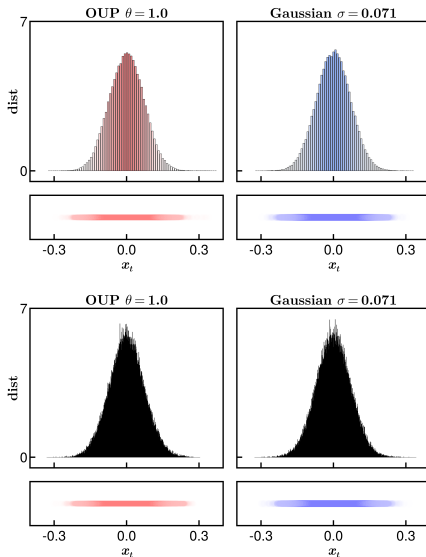
Minimising the *Integrated Mean Square Error* (IMSE) between $\hat{p}(x)$ and $p(x)$

$$\int \mathbb{E}^2[p - \hat{p}]dx \approx \frac{1}{N \cdot h} + \frac{h^2}{12} \int (p')^2 dx,$$

w.r.t. the bin width h yields the optimal number of (uniformly distributed) bins

$$n_* = \left\lceil \text{range}(X_t) / \sigma_s \sqrt{24\sqrt{\pi}N^{-\frac{1}{3}}} \right\rceil,$$

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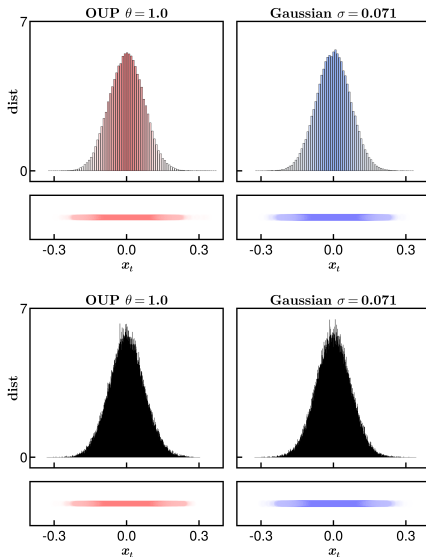
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We then transform the minimisation problem into a weighted one

$$\mathbf{c}^* = \operatorname{argmin}_{\mathbf{c} \in \mathbb{R}^3} \left\{ \sum_{j=1}^{n_*} w_j (p(x_j, \mathbf{c}) - y_j)^2 \right\},$$

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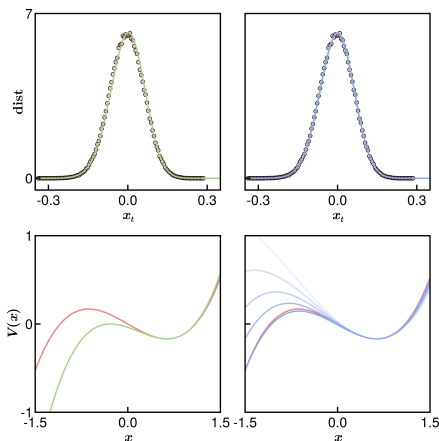
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First attempts at stabilizing the method

